

Rushil Shah

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SKILLS

Languages: Golang, Python, Java, C++, Ruby, TypeScript, JavaScript, Elixir, HTML5, CSS

Technologies: LLM, LangGraph, RabbitMQ, Kafka, React, Django, ElasticMQ, REST, InfluxDB, AWS, GraphQL

Databases: PostgreSQL, Elasticsearch, MongoDB, DynamoDB, Firebase, Cassandra, Redis

PROFESSIONAL EXPERIENCE

Tesla | Software Development Engineer

Feb 2024 - Present

- Converted a WebSocket-based **AI chatbot** integration to **HTTP streaming**, enhancing **scalability**, **reliability**, and **real-time performance** by reducing connection overhead and migrating to the **Orca infrastructure**
- Migrated test-drive survey pipeline from Qualtrics to Tesla's **ECHO** platform and built "TD Hot Leads", converting survey signals (purchase intent, satisfaction) into prioritized high-intent leads surfaced to advisors
- Increased sales leads by **20%** by onboarding SMP as an MCP agent on Tesla's Nova AI platform - driving agent template deployment, OBO auth, ARB security review, and fixing LLM routing and hallucination issues
- Designed an automated inbound-call-to-lead pipeline integrating Avaya telephony with SMS, auto-creating sales order/activities and routing them to the store; the customer called - piloted and rolled across US, NL, and ES
- Designed **Autosweep** to clear out old data based on business requirements, saving **33%** of Tesla Advisors' time via **Goroutines** and **Cron Jobs**, configurable via **SQL** query without production releases for future changes
- Engineered a real-time email notification feature by programming cron jobs to poll data at regular intervals and asynchronously send the email payload to the necessary **RabbitMQ** queues for message transfer.
- Accomplished a **40%** reduction in write errors caused by version conflicts while writing documents in Elasticsearch by implementing a retry mechanism with error handling, random jitters, and async programming using **Goroutines**.
- Enabled storing **database transactions** in the request context to avoid creating multiple transactions for every database call using the **Golang Context** package, improving CPU consumption by **20%**

Fetch | Backend Engineer

Nov 2023 - Feb 2024

- Scaled microservices that scan millions of user receipts by implementing pipelines in **Golang** to check validations, item expiry, and normalization, storing data in the **Snowflake** data lake.
- Led the migration of **314 GB** of data from **AWS ElastiCache** to a compact version of **DynamoDB**, optimizing CPU utilization from **95%** to **32%** through effective fallback and feature flag implementation.

InfluxData | Software Engineering Intern ([General Marketplace Simulator](#))

May 2022 - Aug 2022

- Designed a **notification service** to simulate the **AWS Marketplace** for testing the onboarding of **70%** of company users, using functional programming in **Elixir Phoenix** to send messages to local queues in **Docker**.
- Conceptualized and implemented the architecture to duplicate **AWS SNS** and **SQS** services locally in **Elixir** by sending notifications at runtime and polling message queues at regular intervals, locally using **ElasticMQ**

Shipmnts | Full Stack Software Engineer

Dec 2020 - Jun 2021

- Improved scalability by **30%** for a subscription service that delivers tracking data in real-time by using a **queuing architecture** in **Ruby on Rails** to delegate large asynchronous tasks to different queues for information retrieval.
- Achieved a **50%** reduction in API consumption in a subscription service by programming **cron jobs** in **JavaScript** to fetch and update data asynchronously, and adjusting cron time intervals at runtime based on previous tracking data.

EDUCATION

Master of Science in Software Engineering | San Jose State University, California - GPA: 3.81/4

Aug 2023

PROJECTS

Vini: AI-Driven Chatbot ([Vini](#)) Python | Grok | Vector DB | FastAPI

Feb 2023 - May 2023

- Designed and implemented a personalized chatbot using **LLMs** and vector search to deliver context-aware responses about my education, work experience, and skills, ensuring accurate and dynamic information retrieval.
- Optimized embeddings and retrieval flow, improving response relevance and query accuracy by **40%**, demonstrating expertise in **AI-driven** personalization and natural language understanding.